

School of Computer Science and Artificial Intelligence

Lab Assignment # 7.2

Program : B. Tech (CSE)

Specialization : AIML

Course Title : AI Assisted

Coding Course Code:

23CS002PC304

Semester : VI

Academic Session : 2025-2026

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Date : 03/02/26

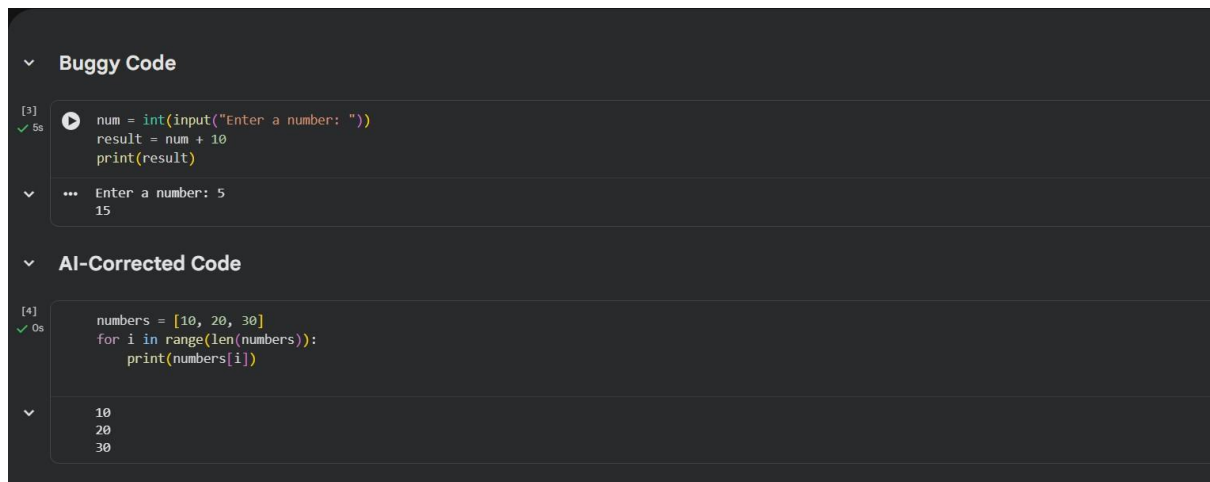
Lab 7: Error Debugging with AI (Week 4 – Tuesday)

Topic: Systematic approaches to finding and fixing bugs using AI

Task 1 – Runtime Error Due to Invalid Input Type

Bug Analysis (AI Explanation)

- `input()` always returns a **string**
- Adding a string and an integer causes a **TypeError**



The screenshot shows a code editor with two sections. The first section, titled 'Buggy Code', contains a Python script where a user input is converted to an integer and then added to 10. The second section, titled 'AI-Corrected Code', shows a different script where a list of numbers is iterated over and each number is printed.

```
[3] 5s num = int(input("Enter a number: "))
result = num + 10
print(result)

... Enter a number: 5
15
```

```
[4] 0s numbers = [10, 20, 30]
for i in range(len(numbers)):
    print(numbers[i])

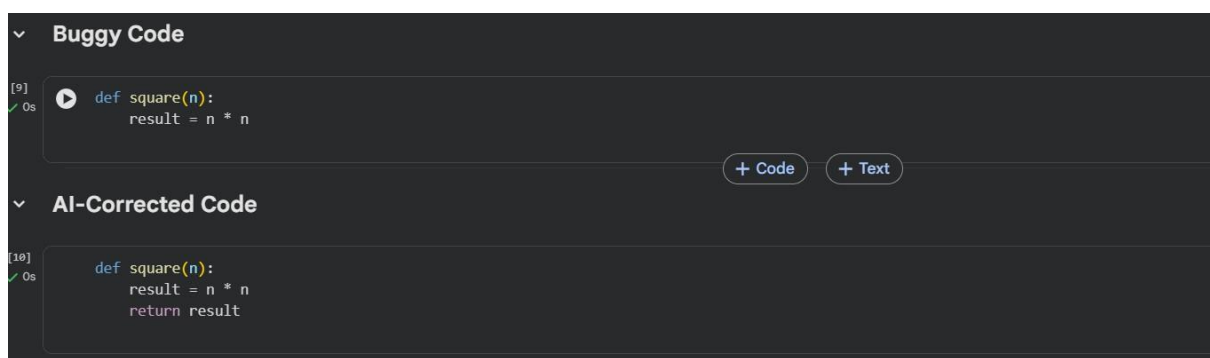
10
20
30
```

Expected Output – 1

- AI converts user input to an integer
- Runtime error is eliminated

Task 2 – Incorrect Function Return Value Bug Analysis (AI Explanation)

- Function calculates the square but **does not return it**
- Without return, Python returns `None`



The screenshot shows a code editor with two sections. The first section, titled 'Buggy Code', contains a Python function that calculates the square of a number but does not return it. The second section, titled 'AI-Corrected Code', shows the same function but with a `return` statement added to return the result.

```
[9] 0s def square(n):
    result = n * n

+ Code + Text
```

```
[10] 0s def square(n):
    result = n * n
    return result
```

Expected Output – 2

- Function correctly returns the square of the number

Task 3 – IndexError in List Traversal Bug Analysis (AI Explanation)

- `range(0, len(numbers)+1)` goes **one step too far**
- Causes `IndexError: list index out of range`

✓ Buggy Code

```
[2] numbers = [10, 20, 30]
0s for i in range(0, len(numbers)+1):
    print(numbers[i])
```

✓ ... 10
20
30

```
-----
IndexError                                Traceback (most recent call last)
/tmp/ipython-input-2172525831.py in <cell line: 0>()
      1 numbers = [10, 20, 30]
      2 for i in range(0, len(numbers)+1):
----> 3     print(numbers[i])

IndexError: list index out of range
```

Next steps: [Explain error](#)

✓ AI-Corrected Code

```
[4] numbers = [10, 20, 30]
0s for i in range(len(numbers)):
    print(numbers[i])
```

✓ ... 10
20
30

[+ Code](#) [+ Text](#)

Expected Output – 3

- Loop boundary corrected
- Prevents out-of-range access

Task 4 – Uninitialized Variable Usage

Bug Analysis (AI Explanation)

- Variable `total` is **used before assignment**
- Causes `NameError`

▼ Buggy Code

```
[5]
0s
if True:
    pass
print(total)
```

▼

```
... -----
NameError                                Traceback (most recent call last)
/tmp/ipython-input-3608487366.py in <cell line: 0>()
      1 if True:
      2     pass
----> 3 print(total)

NameError: name 'total' is not defined
```

Next steps: [Explain error](#)

▼ AI-Corrected Code

```
[6]
✓ 0s
total = 0
if True:
    pass
print(total)
```

▼

```
... 0
```

[+ Code](#) [+ Text](#)

Expected Output – 4

- Variable initialized before use
- Program runs safely

Task 5 – Logical Error in Student Grading System Bug Analysis (AI Explanation)

- Logical order of grading conditions is incorrect
- marks ≥ 80 wrongly assigns grade **C**
- else block assigns **B** incorrectly

Buggy Code

```
[7]
✓ 0s
marks = 85
if marks >= 90:
    grade = "A"
elif marks >= 80:
    grade = "C"
else:
    grade = "B"
print(grade)
```

▼

```
... C
```

▼ AI-Corrected Code

```
✓ AI-Corrected Code

[8] 0s ▶ marks = 85
    if marks >= 90:
        grade = "A"
    elif marks >= 80:
        grade = "B"
    else:
        grade = "C"
    print(grade)

... B
```

Expected Output – 5

- Correct grade is assigned based on marks
- Logical flow fixed

Summary: AI-Assisted Debugging Strategies Used

- ✓ Type conversion for runtime errors
- ✓ Return statement validation
- ✓ Loop boundary correction
- ✓ Variable initialization checks
- ✓ Logical condition reordering